**InvoiceSentinel**

AP RISK DESK · PH SMES

# Stop paying the ghost before the money moves.

A Gemini + Google ADK risk desk that checks every supplier invoice against your vendor master, purchase orders and BIR rules, then tells the analyst **PAY, HOLD or VERIFY**, with evidence.

**Theme · BFSI: Intelligent Risk, Fraud & Financial Experiences**


LIVE PROTOTYPE · CLOUD RUN
invoicesentinel-web-1034883854575.asia-southeast1.run.app

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GOOGLE CLOUD AI BUILDER CUP 2026
Prototype submission · Oct 2026

01 THE PROBLEM

# Fake receipts are an industry, and accounts payable is where they get paid

Problem & impact

## Ghost-receipt cases filed by the BIR (alleged, ₱ billions)

4 ghost corporations: alleged tax loss, 2019–21 [1][2]




Ever Bilena: alleged total tax liability [5]



**₱1.6B**

23 buer firms: alleged deficiencies, 2025 [3][4]



**₱1.41B**

World Balance: alleged basic deficiencies [6]



**₱0.1788B**

**68%** of organizations hit at least one AP fraud attempt in 2024; only **31%** use automated fraud detection (Basware survey of 100 finance leaders [7]).

BIR figures are allegations in complaints and filings, not final court findings. Sources on page 13.

### The documents look real

Ghost receipts carry a TIN, a VAT line and a clean letterhead. They come from corporations with no operations, set up only to issue receipts [3].

### The buyer takes the hit

In the 2025 cases, purchasing companies, their officers and their CPAs faced complaints [3]. A fake expense means cash out the door, then audit exposure.

### Bank-change scams hit the same desk

A "new bank account, please don't call" note on an otherwise normal invoice can empty an SME account in one transfer. Most SMEs have no fraud team; the only control is the clerk who opens the PDF.

02 THE SOLUTION

# A pre-payment risk desk: upload, check, decide

GenAI UX

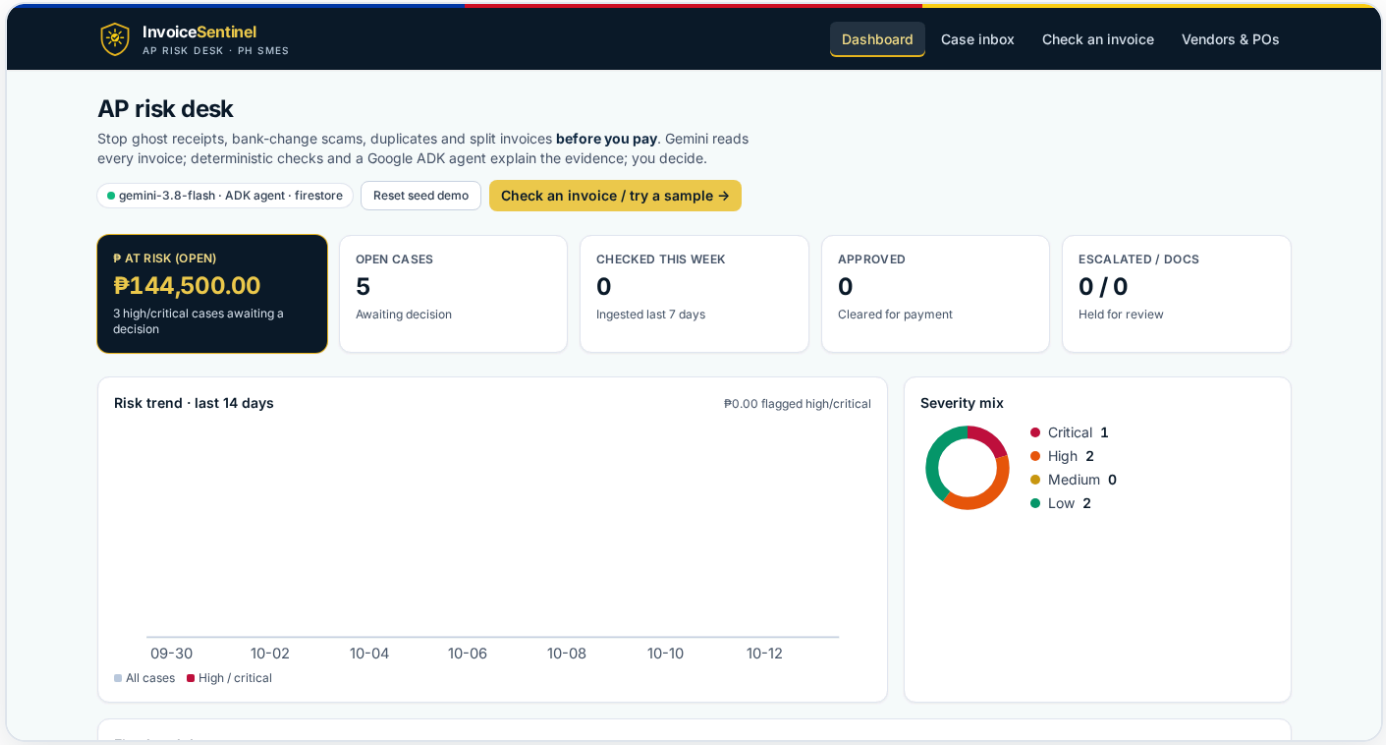
**1 Upload**
Photo, scan or PDF, or one click on **Try a sample**.

**2 Gemini reads**
Fields, line items, payee and bank details in structured JSON. No OCR templates.

**3 Code checks**
Vendor master, PO, TIN, BIR RR 7-2024, split invoices, VAT math: deterministic rules.

**4 Agent explains**
Ask the case in any language; the ADK agent cites evidence and drafts a verification email.

**5 Human decides**
Approve, Escalate or Request docs. Every step goes to the audit trail.




Live dashboard: KPI cards, risk trend, flag breakdown and severity mix.

## Design principles

- **Triage, not verdicts.** The score orders the queue; it never pays or blocks anything by itself.
- **Evidence for every claim.** Risk cards and agent answers cite evidence IDs (E1–E9, IMG) that the analyst can open.
- **Code for facts, AI for judgment.** Matching, TIN checks, BIR rules and scoring are deterministic; Gemini reads the document and explains.
- **Never fake an answer.** If Gemini is down, the API returns a retryable error. It never invents a clean case.
- **Honest scope.** Supports AP triage; it does not certify BIR compliance.

03 THEME FIT

# Built for BFSI: trust, fraud and risk at the last control before cash moves

Problem alignment

### Fraud

Named fraud types are first-class flags: ghost vendor, bank-change / BEC payment redirection, duplicates, overbilling, split invoices that dodge approval limits.

### Risk

A deterministic 0–100 score with severity floors ranks the inbox. Shows "₱ at risk" and high/critical open cases, so controllers can see their exposure at a glance.

### Trust

Every flag is explained with evidence. The verification email goes only to the contact *on file*, never the one printed on the invoice. A person makes every payment decision.

## Financial experience for the people who pay

| USER | GETS |
|---|---|
| **AP clerk** | Ranked inbox, one-click samples, plain-language answers in English, Tagalog and more |
| **Controller** | KPI dashboard, Escalate / Request docs, timestamped audit trail |
| **SME owner** | Protection from bank-change scams for centavos per invoice, on a phone |
| **Banks / lenders** | A pattern that carries over to SME payment portals and claims-document review |

## Why it matters beyond one company

- The theme asks for AI that detects and mitigates fraud or risk and automates complex financial processes [10]. InvoiceSentinel does both, before the payment goes out.
- BIR RR 7-2024 invoice checks and TIN validation make it harder for ghost receipts to enter the books, which protects tax revenue.
- A light browser tool rather than an ERP, so SMEs can adopt it without a fraud team.

# Deterministic checks catch what a tired reviewer misses

Technical merit | Innovation

### Vendor master & PO matching

Vendor, TIN, bank account, PO number, line quantities and prices vs the master. A changed bank account is a **critical** flag.

### TIN validation

Format and branch-code checks; compares the vendor TIN with the master record.

### BIR RR 7-2024 checklist

Required invoice elements with section references, buyer-TIN rule for VAT sales ≥ ₱1,000. Advisory only.

### Split-invoice detection

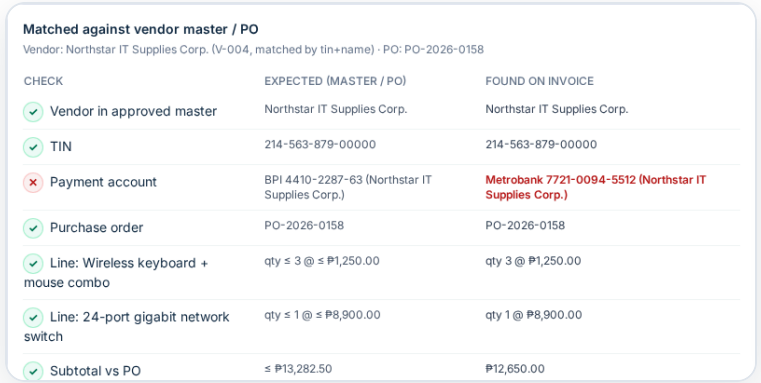
Same vendor, close dates, combined total above the approval limit (set in Settings) → **SPLIT_INVOICE**.

### Deterministic scoring

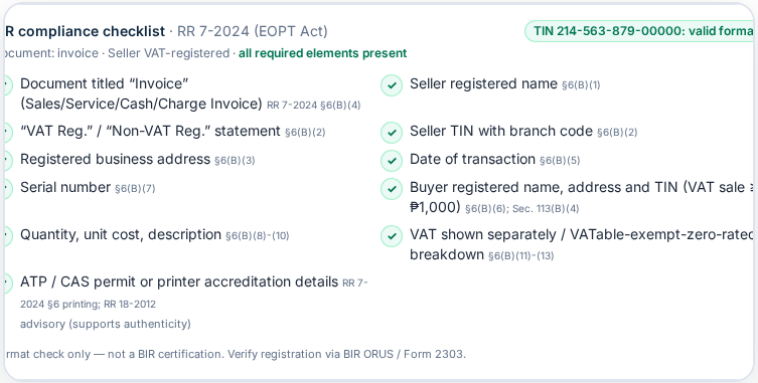
Severity weights (low 20 · medium 45 · high 70 · critical 90) +5 per extra flag. Same input, same score.

### Gemini extraction

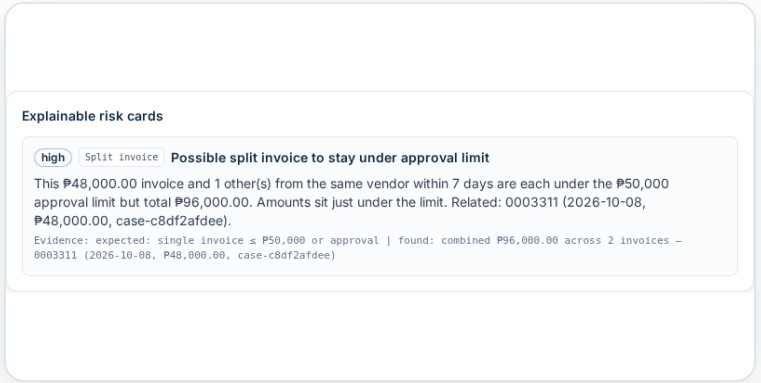
Multimodal read with a JSON schema at temperature 0, seed 7, with retries and a 60 s timeout.




Sample 09: bank account differs from the vendor master.




BIR RR 7-2024 checklist with TIN status.




Sample 13 flagged as a split of sample 12.

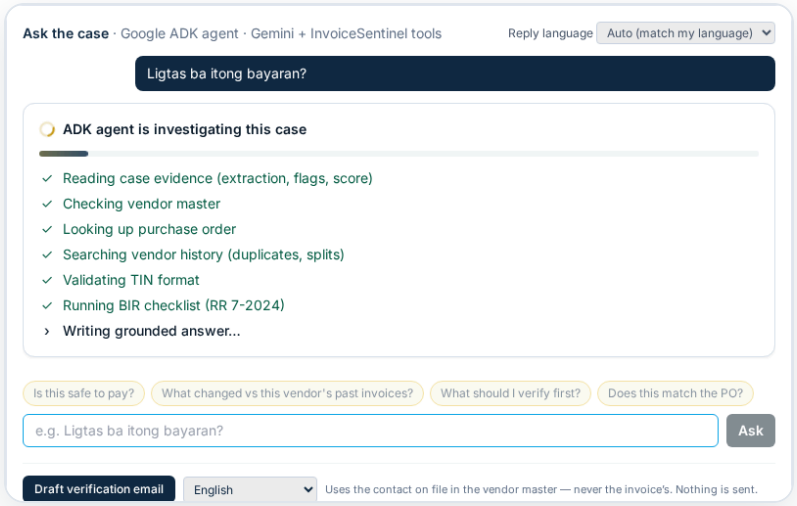
# Ask the case: a Google ADK agent that shows its work as it goes

GenAI UX

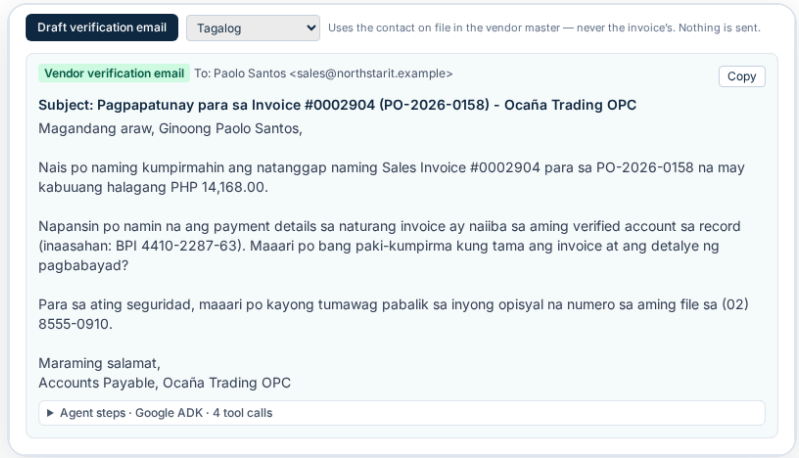
**7 tools, all read-only**

get_case_evidence · lookup_vendor · lookup_po · validate_tin · bir_checklist · search_vendor_history · draft_verification_email

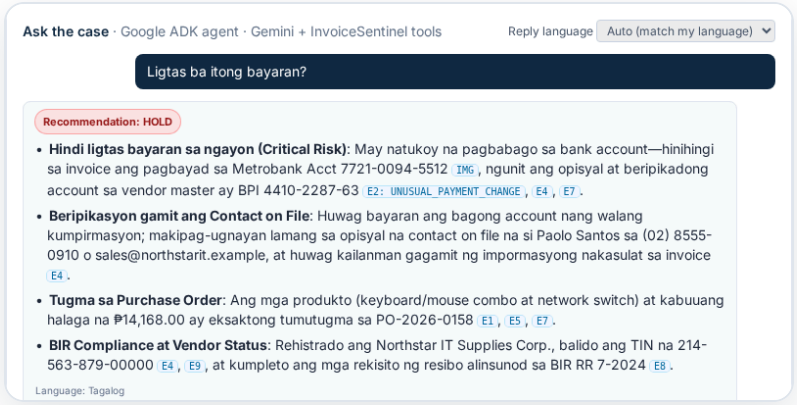
- **Streamed steps:** each tool call shows up live ("Checking vendor master…") over NDJSON. The first step arrives in about 5 s.
- **Grounded:** answers cite evidence IDs and end with **PAY, HOLD or VERIFY**.
- **Multilingual:** replies in the user's language. Tagalog and English were tested live.
- **Safe email:** code, not the model, sets the recipient from the vendor master. Drafts are copied, never auto-sent.




Agent steps arriving live.



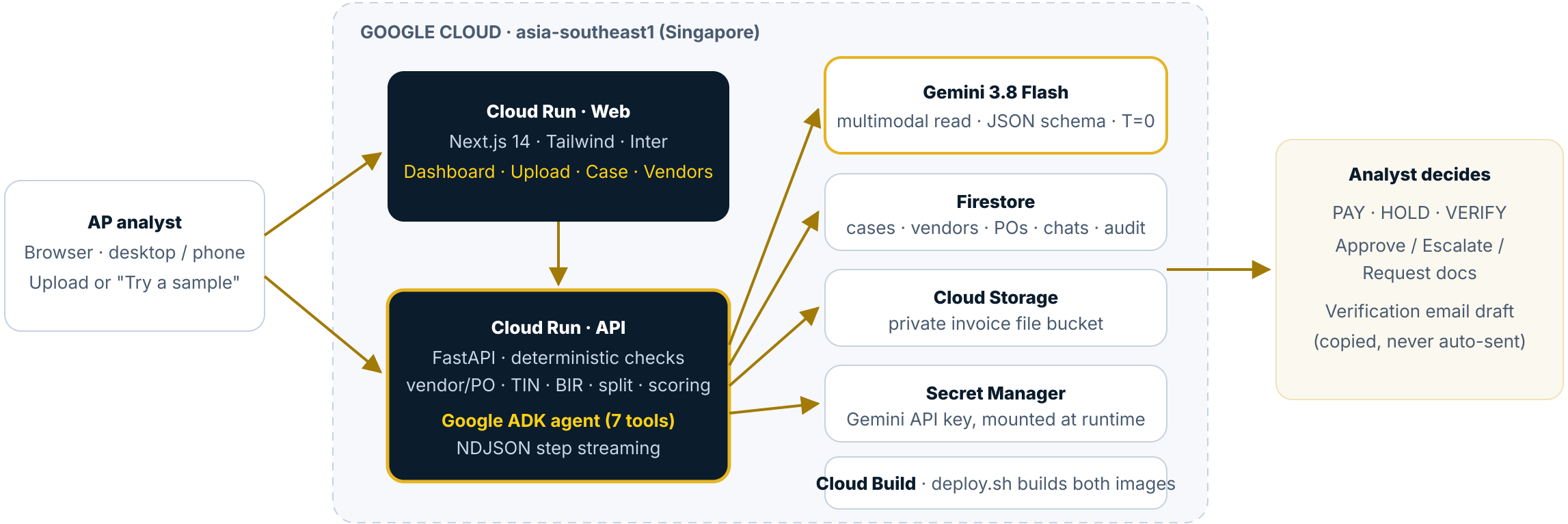

Tagalog verification email to the contact on file (sample 09).




Tagalog answer: HOLD, with cited evidence.

# Serverless on Google Cloud, end to end

Technical merit




**Stateless services.** Both containers autoscale on Cloud Run; all state lives in Firestore and Cloud Storage.

**Secrets stay secret.** The Gemini key is in Secret Manager, never in code or images. Uploads go to a private bucket.

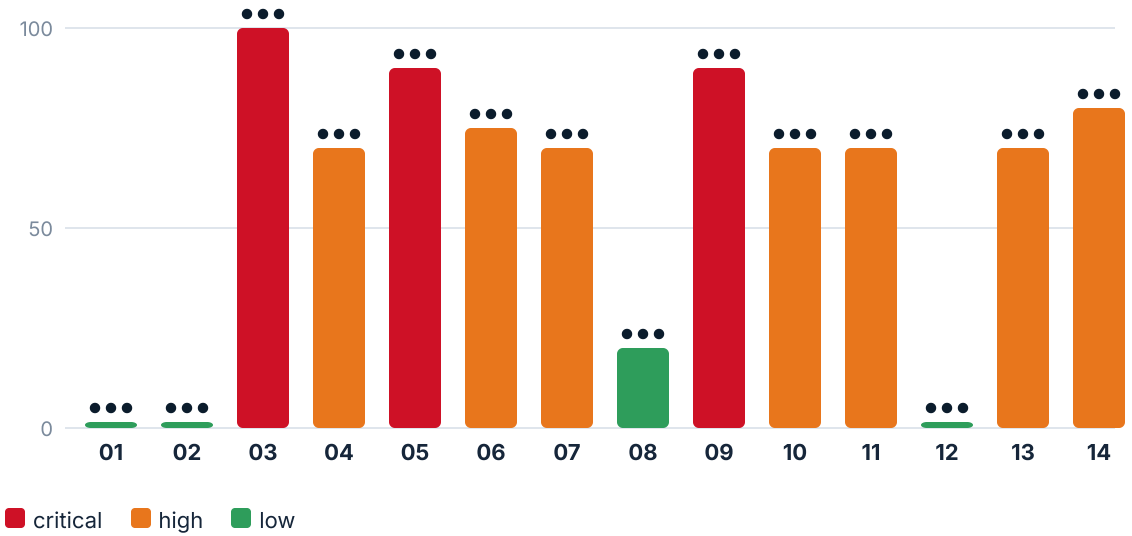
**One-command deploy.** `deploy.sh` provisions Firestore, the bucket and IAM, then builds and deploys both services with Cloud Build.

07 EVALUATION

# 14 test invoices × 3 runs: identical results every time

Technical merit

## Risk score per sample (dots = 3 runs, all equal)




**0**
score spread over 3 passes, all 14 samples

**14**/14
in the expected severity band

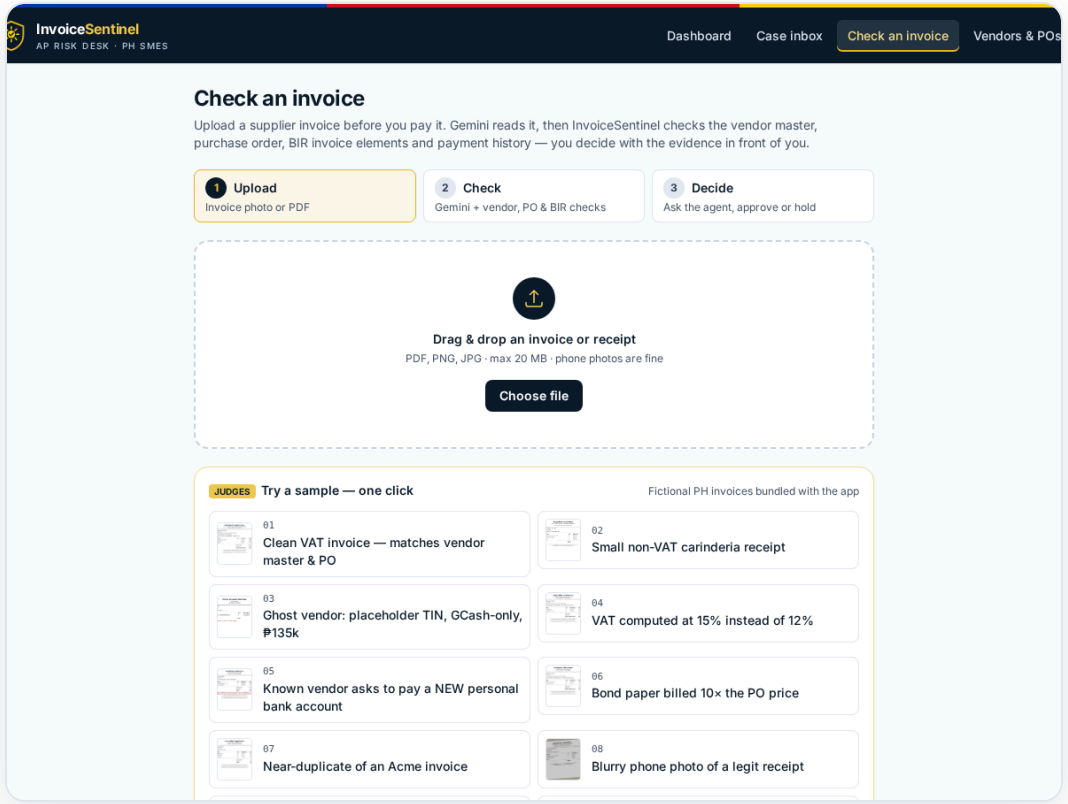
**~20**s
upload to verdict, live on Cloud Run

| # | SAMPLE (FICTIONAL) | BAND | SCORES | KEY FLAGS |
|---|---|---|---|---|
| 01 | Legit VAT invoice | low | 0 / 0 / 0 | — |
| 02 | Small non-VAT receipt | low | 0 / 0 / 0 | — |
| 03 | Ghost vendor | critical | 100 / 100 / 100 | GHOST_VENDOR, TIN_INVALID |
| 04 | VAT math wrong | high | 70 / 70 / 70 | MATH_ERROR |
| 05 | Bank-change BEC | critical | 90 / 90 / 90 | UNUSUAL_PAYMENT_CHANGE |
| 06 | Overbilled vs PO | high | 75 / 75 / 75 | OVERBILLING, PO_MISMATCH |
| 07 | Near-duplicate | high | 70 / 70 / 70 | DUPLICATE_INVOICE |
| 08 | Blurry legit photo | low | 20 / 20 / 20 | VENDOR_TIN_MISMATCH (low) |
| 09 | Known vendor, new bank | critical | 90 / 90 / 90 | UNUSUAL_PAYMENT_CHANGE |
| 10 | Qty exceeds PO | high | 70 / 70 / 70 | PO_MISMATCH |
| 11 | PO not found | high | 70 / 70 / 70 | MISSING_PO |
| 12 | Split invoice A | low | 0 / 0 / 0 | — |
| 13 | Split invoice B | high | 70 / 70 / 70 | SPLIT_INVOICE |
| 14 | Malformed TIN, no BIR fields | high | 80 / 80 / 80 | TIN_INVALID, BIR_FORMAT_ISSUE |

Run with `samples/stability.py 3` on gemini-3.8-flash, Oct 12, 2026, resetting between passes. Bank-change samples 05 and 09 and split samples 12 and 13 were re-checked on the live deployment on Oct 13. Sample 08 is a known limit: a blurry TIN gives a low-severity flag, not a fraud flag.

# Try it in 60 seconds, with no files needed

UX

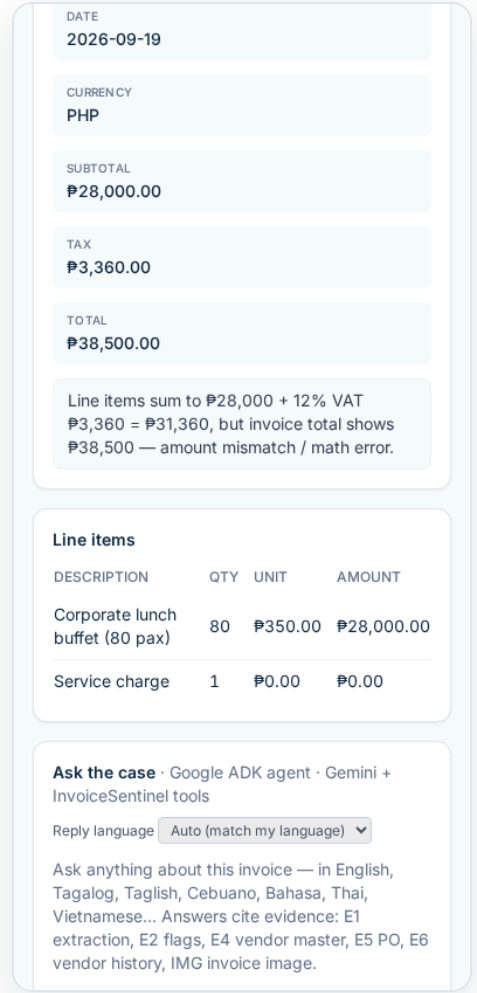



Upload page: stepper, drop zone and the **Try a sample** grid.

**LIVE URL**
**https://invoicesentinel-web-1034883854575.asia-southeast1.run.app**

1. **Check an invoice → Try a sample.** Click **09 known vendor, new bank**. The steps animate while Gemini reads it.
2. **Read the case.** A critical risk card plus a vendor-master table showing the bank-account mismatch.
3. **Ask the case.** Type "Ligtas ba itong bayaran?" and watch the agent steps stream in.
4. **Draft verification email** in Tagalog. It goes to the contact on file. Copy it; nothing is sent.
5. **Splits:** run sample 12, then 13. The second one is flagged SPLIT_INVOICE.
6. **Decide:** Approve, Escalate or Request docs. The decision is logged in the audit trail.
7. **Vendors & POs** page: edit the master data the checks use.

"Reset demo" restores the seed data. All vendors, people and accounts are fictional.




Mobile layout.

09 COMMUNITY BENEFIT & ACCESSIBILITY

# Fraud protection for the businesses that have no fraud team

Impact UX

### SMEs

MSMEs are most businesses in the Philippines, and most have no AP controls beyond one person. A browser tool that works from a phone photo brings bank-grade checks within reach.

### Tax revenue

Every ghost receipt that is stopped is a fake deduction and fake input VAT kept out of the books. BIR RR 7-2024 checks point clerks to the missing elements before payment.

### Multilingual

Clerks ask in Tagalog, English or mixed Taglish and get answers and supplier emails in that language, so staff and suppliers both understand them.

### Accessibility built in

- WCAG AA contrast for body text; severity is shown as text labels, not only color
- Visible gold focus rings and full keyboard navigation; aria-current in the nav
- ARIA labels on inputs, live regions for agent progress, reduced-motion support
- Responsive layouts tested at phone width (390 px)

### Responsible by design

- Human in the loop: no automatic rejection or payment
- Small and non-VAT vendors are not penalized for normal informality; blur is not treated as fraud
- Emails are only drafted; the recipient comes from the master record
- Paid-tier Gemini API: content is not used to train Google's models [8]

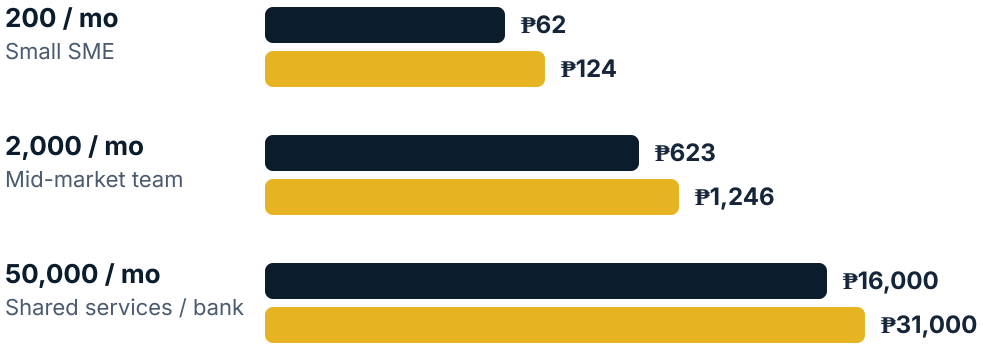
# Centavos per invoice, from a single SME to a bank portfolio

Scalability · Impact

**~₱0.31**

Gemini cost per invoice (3.8 Flash, promo pricing to Dec 31, 2026; ~₱0.62 from 2027) [8]

## Monthly model cost by volume




■ 2026 promo · ■ from 2027. FX ₱62.77/US$. Log scale.

## Infrastructure

- Cloud Run scales to zero; at SME volumes usage fits within the free tier [9].
- For judging, one instance of each service stays warm (min-instances 1) so there are no cold starts: about **US$23 / ~₱1,450 per month**. Set to 0 afterwards.
- Firestore and Cloud Storage cost almost nothing at prototype scale.

## Scalability

- Stateless API, horizontal autoscaling
- Gemini Batch API for overnight payment runs at half the token price [8]
- Tenant-scoped Firestore collections for multiple companies

## Path to production

| | |
|---|---|
| **Done** | Cloud Run, Firestore, private bucket, Secret Manager, ADK agent, deterministic controls |
| **Next** | Identity Platform sign-in with analyst / approver roles; signed URLs; rate limits |
| **Then** | Per-tenant data isolation, retention policies, upload malware scanning, audit export |
| **Ops** | Budget alerts, labeled evaluation set and regression tests on every prompt or model change |

11 ROADMAP

# From prototype to the payment-control layer for PH SMEs

Innovation Impact

**NOW · OCT 2026**

## Live prototype

- Vendor/PO matching, TIN, BIR RR 7-2024, split detection
- ADK agent with streamed steps
- Multilingual Q&A and verification emails
- Firestore + Cloud Storage on Cloud Run

**Q1 2027**

## Pilot-ready

- Sign-in and approver roles
- Email-inbox intake of supplier bills
- CSV import of vendor master and POs
- Labeled PH evaluation set, precision/recall per flag

**Q2 2027**

## Integrations

- QuickBooks / Xero bill sync; hold payment on Escalate
- Batch review of whole payment runs
- Bank-account change history per vendor

**LATER**

## Ecosystem

- Bank SME portals: pre-transfer check API
- Claims-document review for insurers
- Tagalog / Cebuano UI
- Optional BIR ATP and series lookups as signals

**Guardrails stay fixed as we grow:** a person decides every payment, every flag carries evidence, nothing is auto-sent, and the product never claims to certify BIR compliance.

12 TEAM & REFERENCES

# The team behind InvoiceSentinel

Submission




TEAM

## BKD KUBO

Built in the Philippines for Philippine SMEs.

**Manuel III Ocaña**
Product Lead & Solution Architect

**Elson Mojica**
Domain Lead (Finance) & QA

### SUBMISSION LINKS

| | |
|---|---|
| Live prototype | invoicesentinel-web-1034883854575.asia-southeast1.run.app |
| Demo video | [VIDEO LINK] |
| Theme | BFSI: Intelligent Risk, Fraud & Financial Experiences |

## References (accessed Oct 12, 2026)


1. Rappler, "P25.5 billion 'lost': BIR files tax complaints vs firms selling fake receipts," Mar 16, 2023.
2. BusinessWorld, "BIR goes after firms issuing fake receipts," Mar 17, 2023 (₱17.63B income tax, ₱7.91B VAT).
3. BIR press release, Aug 7, 2025: 23 corporations, 56 officers, 17 CPAs; ₱1.41B. bir-cdn.bir.gov.ph/BIR/pdf/PR37AUG0725.pdf
4. Philstar, "23 firms face tax evasion cases for P1.4 billion ghost receipts," Aug 8, 2025.
5. Philstar, "BIR files case vs Ever Bilena over P1.6 billion tax liability," Feb 6, 2024.
6. Manila Bulletin, "BIR snares World Balance over fictitious receipts," Jul 21, 2025 (₱178.8M).
7. Basware / sharedserviceslink, "The Rise in AP Fraud: Frequency, Sophistication, and Impact."
8. Google AI for Developers, "Gemini Developer API pricing," updated Oct 9, 2026. ai.google.dev/gemini-api/docs/pricing
9. Google Cloud, "Cloud Run pricing." cloud.google.com/run/pricing
10. Google Cloud AI Builder Cup 2026, themes and judging criteria. aibuildercup.com/themes.html
11. BIR Revenue Regulations No. 7-2024 (invoicing requirements under the EOPT Act).